Introduction Manual

S400 Dual Laser System Laser Engraver

The S400 Dual Laser System Laser Engraver is used to cut / engrave a variety of materials into desired shape / pattern.
Basic information of the S400:

- Application: material cutting / engraving
- Material: non-metal (e.g. acrylic, plywood, fabric, leather, fur, paper, etc.) & metal (engraving only and using Dual Laser only)
  (Note: Consultation should be done prior operation if materials not listed above will be used.)
- Max working area: 1016 x 610 mm
- Max working speed: 140 inches/second
- Max thickness: acrylic 10 mm
- Compatible data format: DXF, CDR & AI

Pre-cutting:

A. Register for a Laser Machine

Register before use! Register before use! Register before use!
Otherwise, NO QUOTAS will be offered!

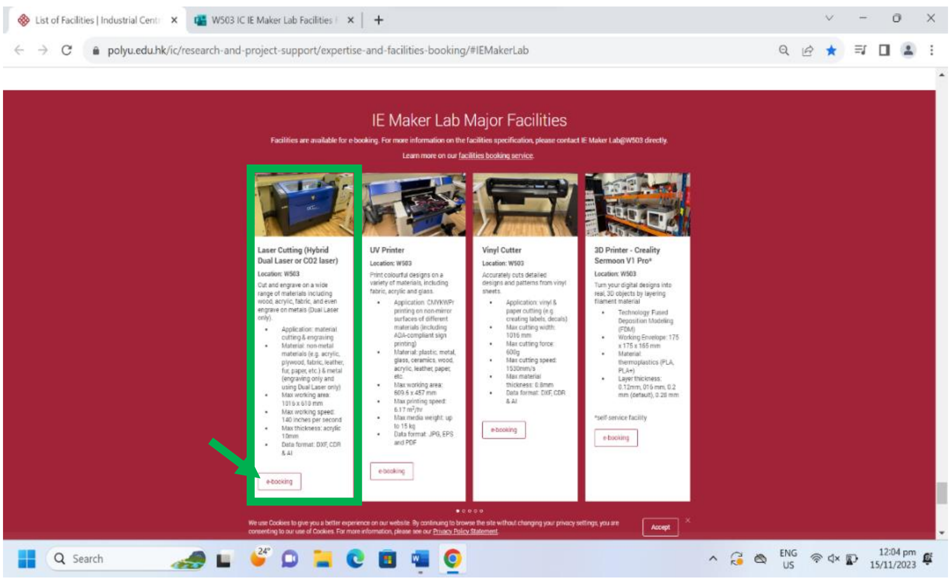
Register through the 2 ways below:

QR code:

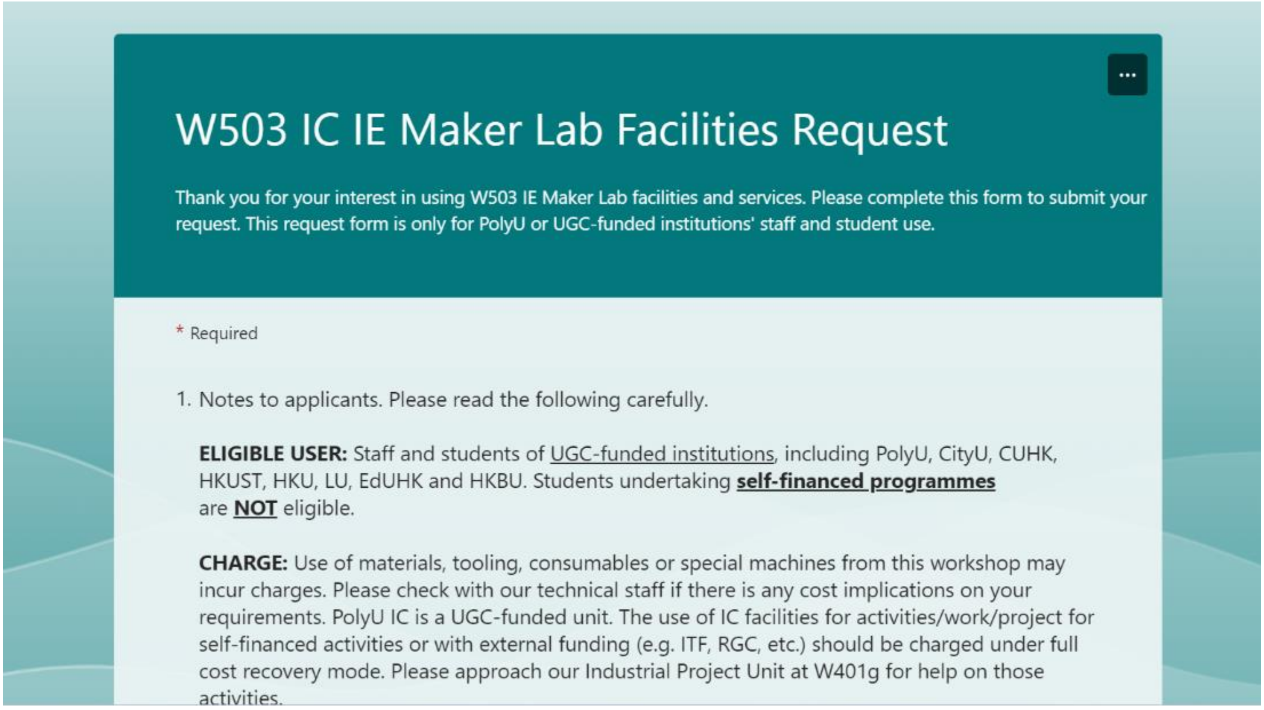
Link:
https://forms.office.com/pages/responsepage.aspx?id=YPc3j4exgUm4L-Xcb1vPRJJiz2e83RZCgJc8RvkhNRRUQVFGS0NMSzRYUVVQWTQ0UVNZSzhBTDhWTS4u

Steps to register:

On the IC website, go to ***List of Facilities > IE maker lab > Laser Cutting (Hybrid Dual Laser or CO2 laser) >*** click ***e-booking***



Fill in the “W503 IC IE Maker Lab Facilities Request” form



W503 IC IE Maker Lab Facilities Request

Thank you for your interest in using W503 IE Maker Lab facilities and services. Please complete this form to submit your request. This request form is only for PolyU or UGC-funded institutions' staff and student use.

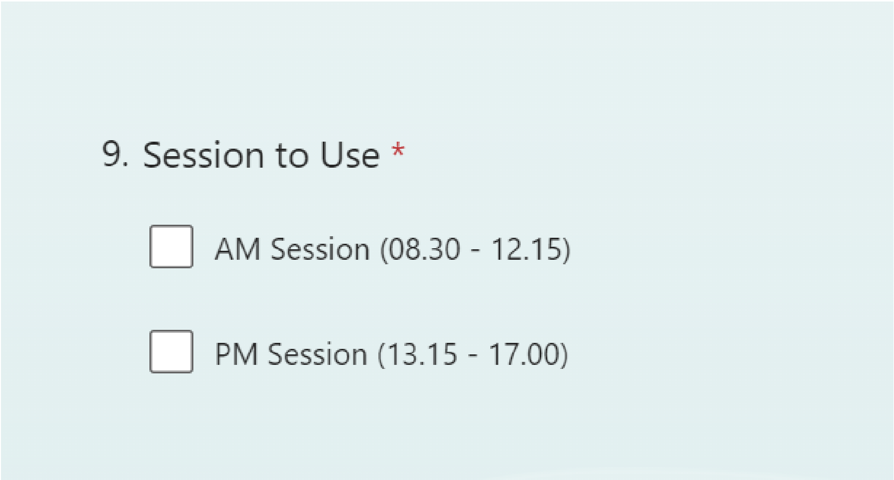
* Required

1. Notes to applicants. Please read the following carefully.

ELIGIBLE USER: Staff and students of UGC-funded institutions, including PolyU, CityU, CUHK, HKUST, HKU, LU, EdUHK and HKBU. Students undertaking **self-financed programmes** are **NOT** eligible.

CHARGE: Use of materials, tooling, consumables or special machines from this workshop may incur charges. Please check with our technical staff if there is any cost implications on your requirements. PolyU IC is a UGC-funded unit. The use of IC facilities for activities/work/project for self-financed activities or with external funding (e.g. ITF, RGC, etc.) should be charged under full cost recovery mode. Please approach our Industrial Project Unit at W401g for help on those activities.

Remember to select a desired time slot.



9. Session to Use *

- [ ] AM Session (08.30 - 12.15)
- [ ] PM Session (13.15 - 17.00)

(Important! If the machine was not admitted by the applied student within 15 mins from the beginning of the time slot, it will be offered to other students.)

B. Confirm booking

Step 1. Present your "Confirm Booking email"

An email will be sent to you if the booking is successful. Please present it to our staff / student helpers to confirm the booking.

Step 2. Select a desired Machine to operate

The two S400s generally offers same functions. They can be used to cut *acrylic, wood, textile and leather*. However, the one on the right provides *a unique Metal engraving function.* Choose the one that fits your job requirement to use.

Step 3. Get the key for the S400

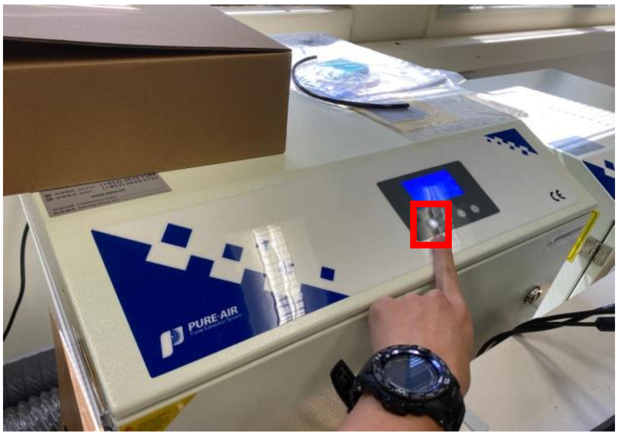
Our staff / student helpers will give you the key to activate the S400.

C. Things to do with the Laser Machine

Complete all steps in this section before moving on!!!

Step 1. Turn on the ventilation

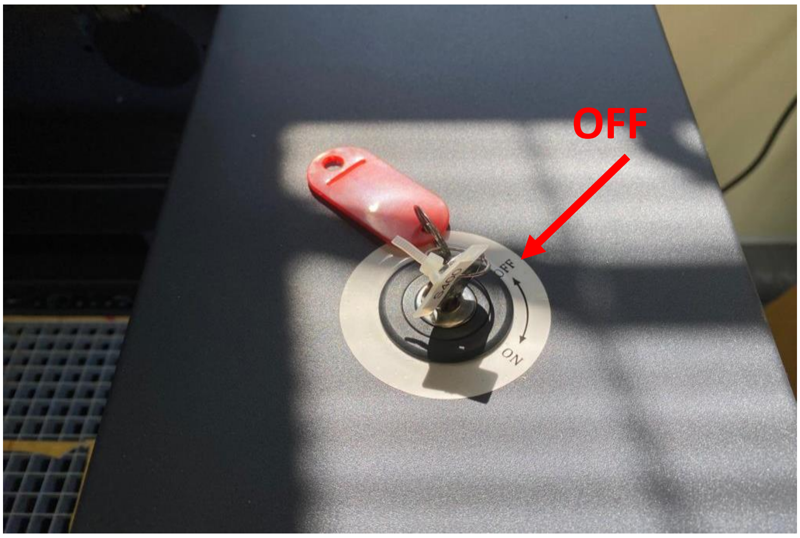
Switch on the ventilators by tapping on the “on/off” button shown below.



Step 2. Turn on the machine

Switch on the machine and insert the key. Keep the key in ***“OFF”*** position.

The machine is going to initialize itself now.



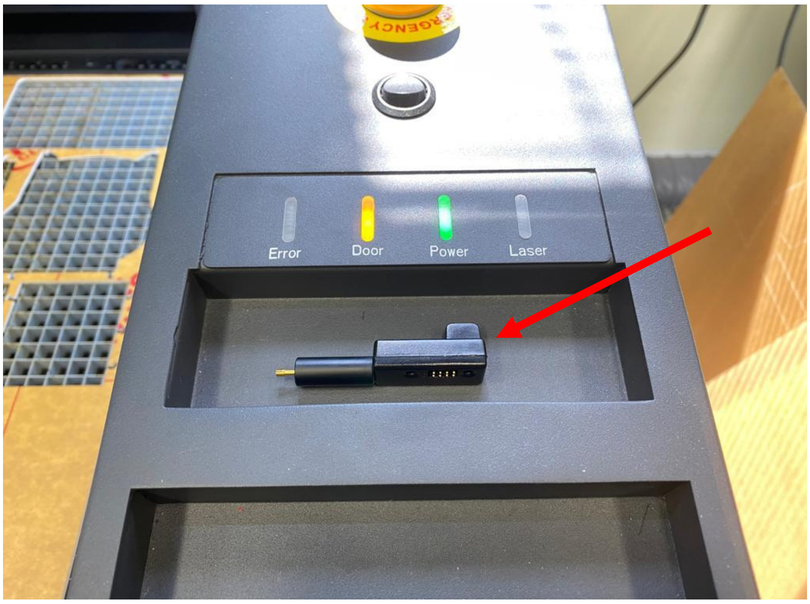
Step 3. Place the material properly

Make sure the material is placed within the work table.

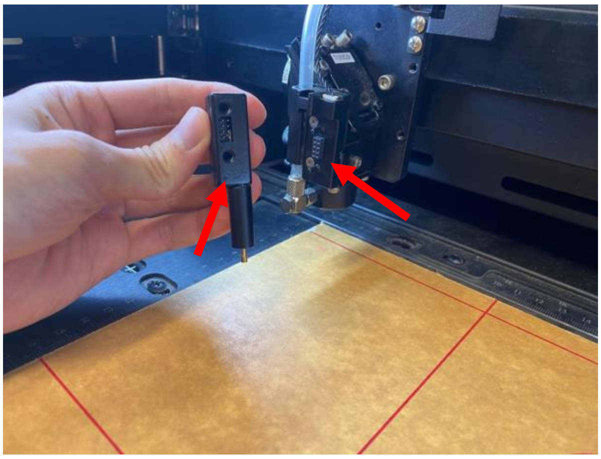
Step 4. Auto focus

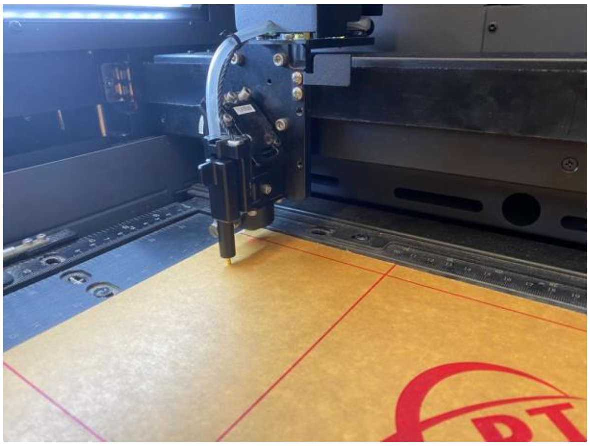
This step is NECESSARY when the cutting materials are changed. (e.g. 3mm acrylic > 2mm acrylic)

4.1. Find the auto focus tool on the machine.

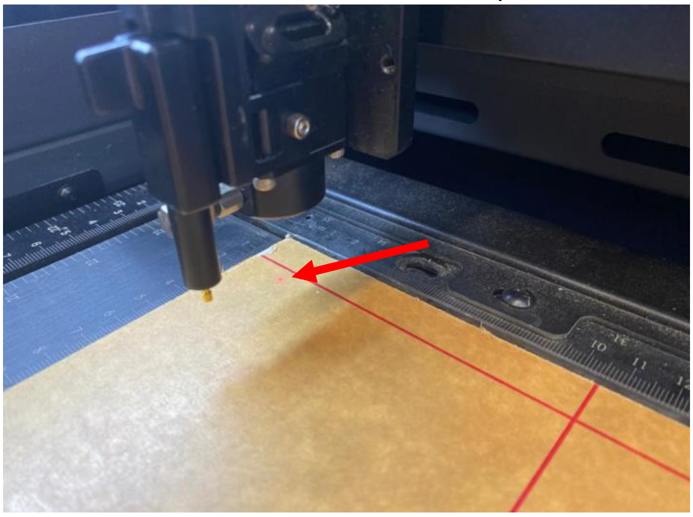


4.2. Insert the tool onto the laser head.





4.3. Move the laser hand and locate the laser pointer on the material.

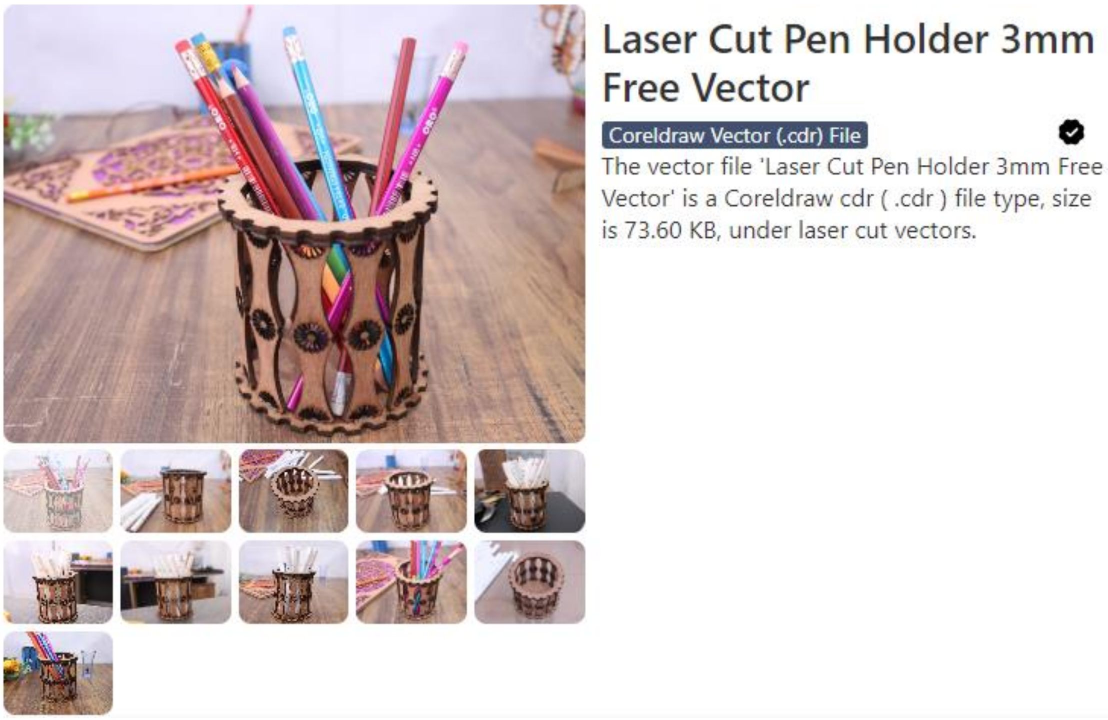


4.4. Tap ***"Auto Focus"***

The tool will tap on the material once to measure its thickness. **After finishing, remove the tool and put it back.**

D. Things to do on PC

Step 1. Download a file on the Internet.



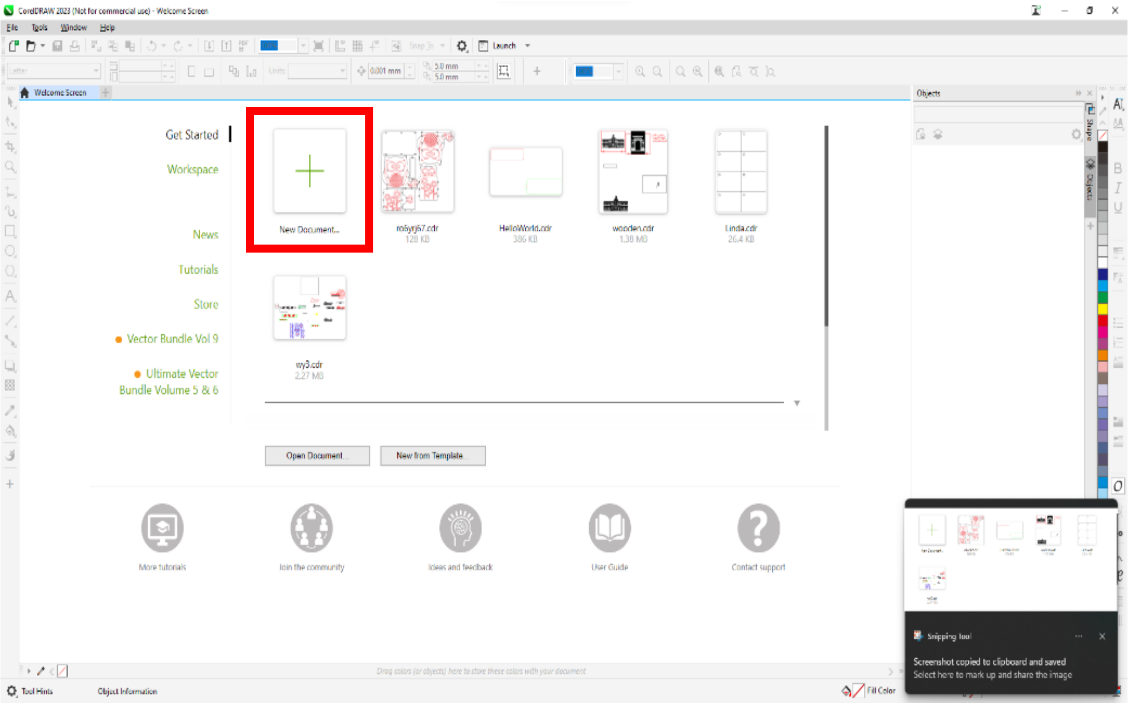
These websites provide laser cut files (DXF, CDR & AI) compatible with CorelDRAW.
3axis.co: https://3axis.co/
Thingiverse: https://www.thingiverse.com/

Step 2. Open CorelDraw

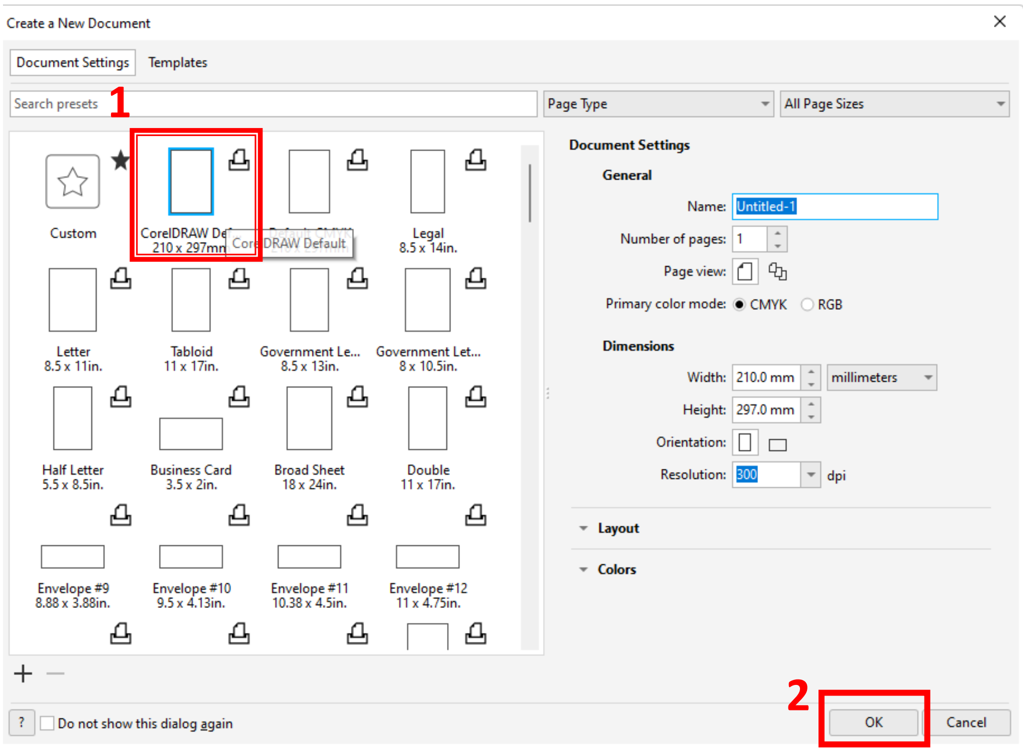
Click on the CorelDRAW icon on the home screen to open the app.

Step 3. Prepare the template

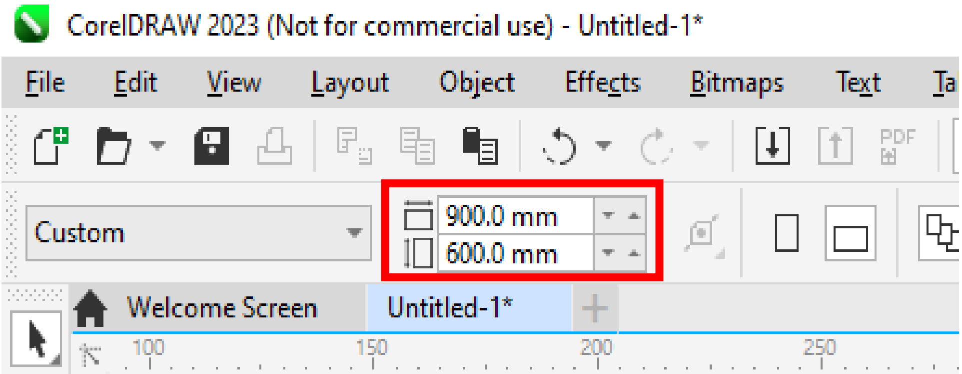
3.1. choose “new document”



3.2. Choose “210 x 297mm (A4)” > “OK”.

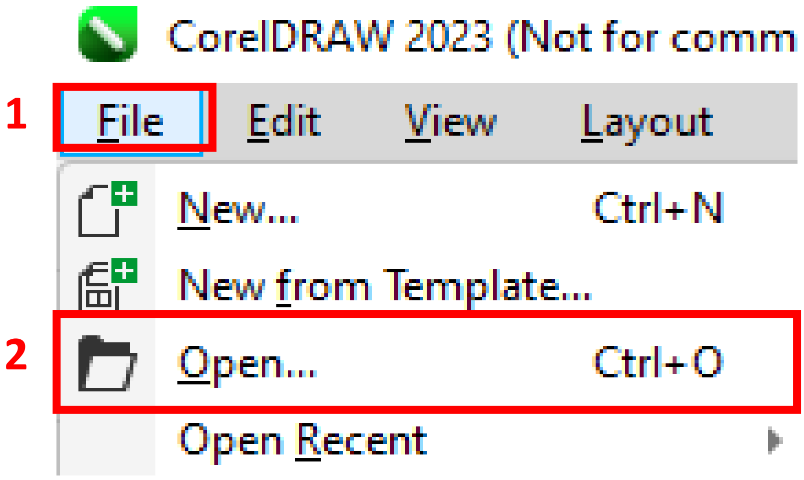


3.3. Set frame size to *900 x 600mm.* This is the size of the laser cutter's work table.

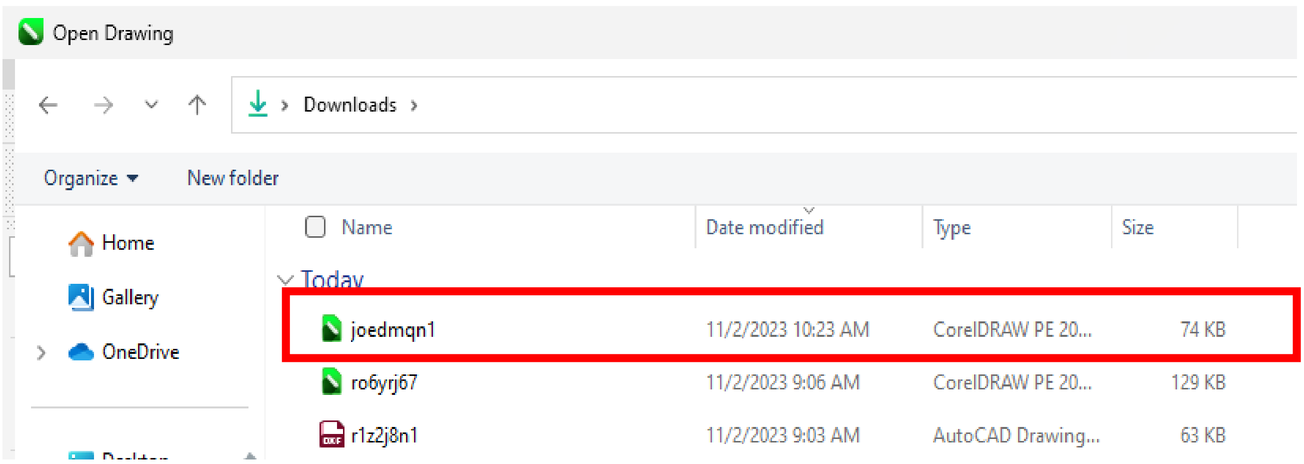


Step 4. Open the downloaded file

Click ***"File > Open"***



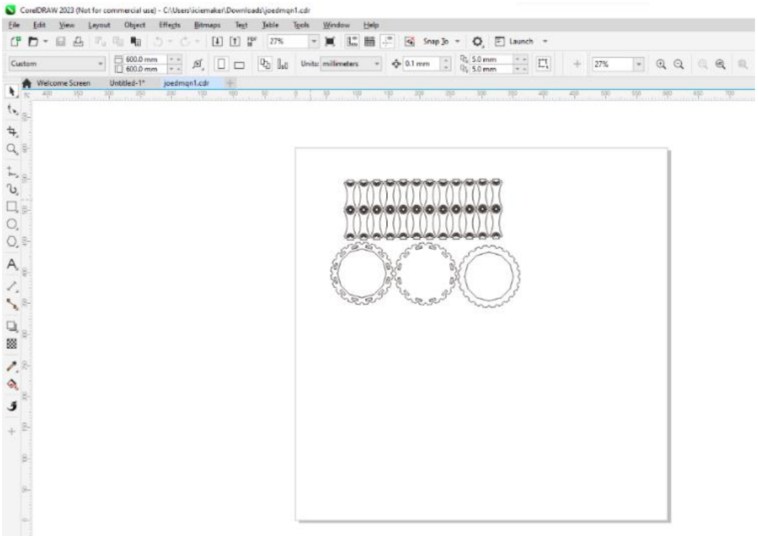
or simply drag it into CorelDRAW from the file explorer.



Step 5. Arrange the objects

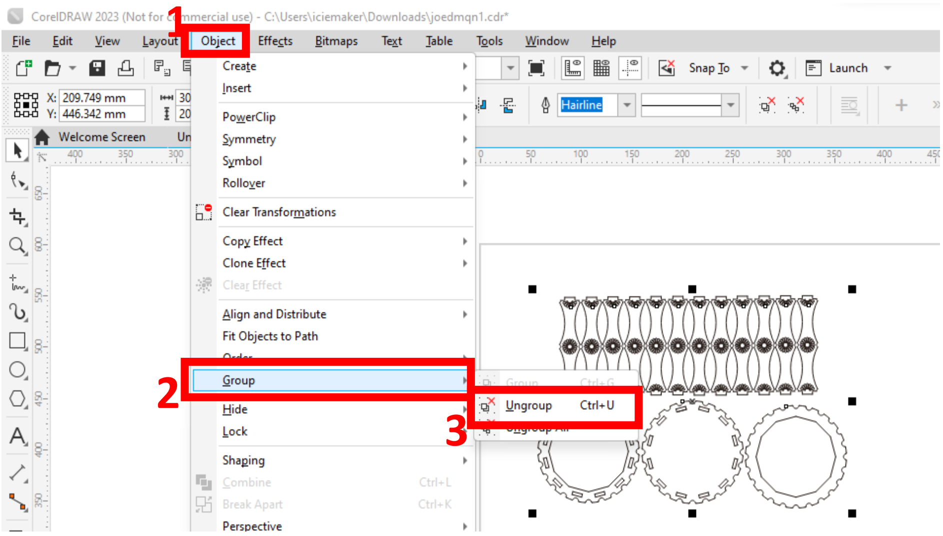
5.1. Ungrouping objects

(Note: some files may have the objects grouped together. You cannot process each object individually. In this context, ungrouping them is necessary. If the objects are NOT grouped, this step can be skipped.)

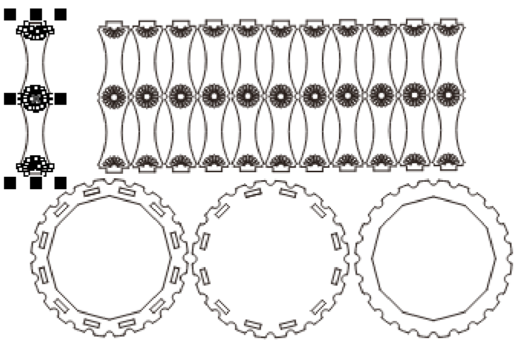


Choose the object. Then go to "***Object> Group> Ungroup***".

(Note: **DON'T choose "Ungroup all"** otherwise all the line segments will be separated.)

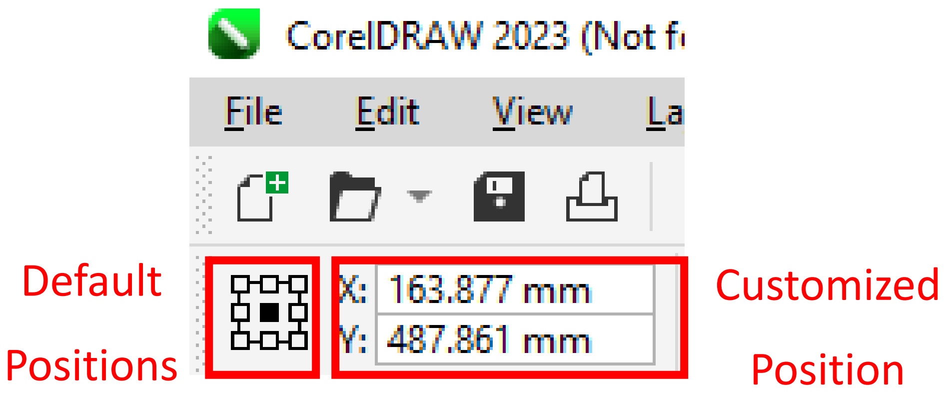


Now you can separate the objects.

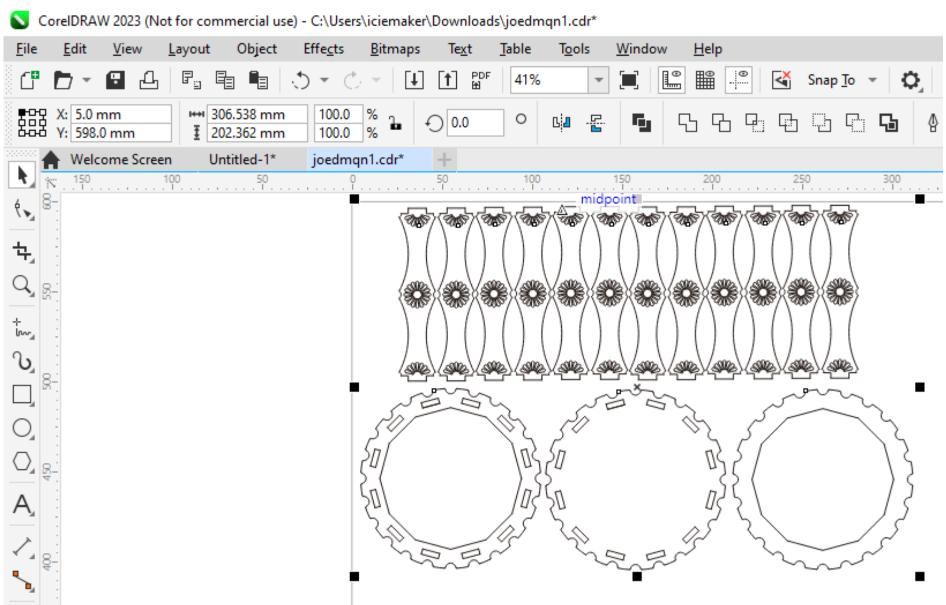


5.2. Place the object on the template properly.

Place the object on 9 default positions on the template by clicking the 9 boxes on the left, or choose a desired position by entering (x, y) coordinates on the right. (Note: the (0, 0) position represents the Bottom left corner)



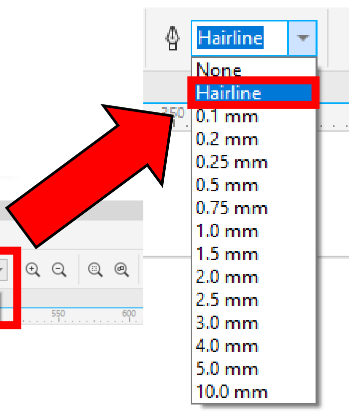
This is an example.

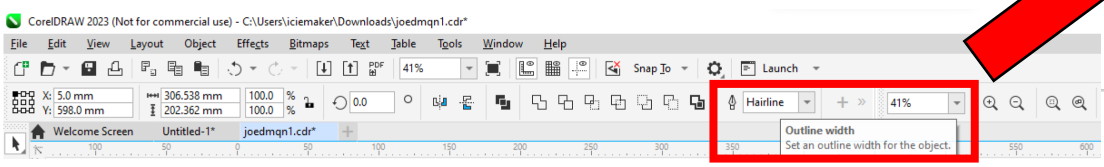


Step 6. Checking the settings

6.1. Line width

At the menu bar, go to "***Outline width***".





You will see a variety of line width settings.

For the laser cutters in W503, ***"Hairline"*** is for cutting, while the others are for engraving.

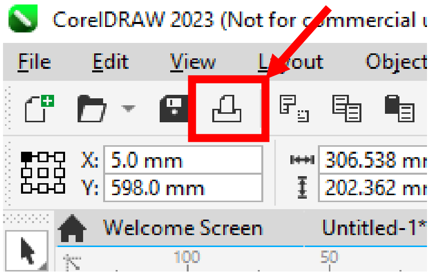
Assign the lines with the suitable one.

(Note: this menu only demonstrates cutting, so "Hairline" is selected.)

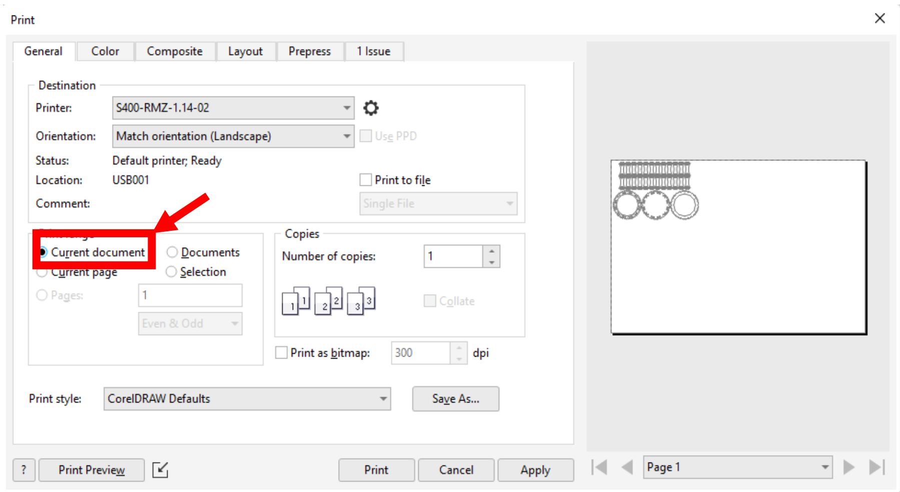
6.2. Print Range

Sometimes we may want to cut/engrave everything in the file, but sometimes we may only want to cut a specific object. The instructions below are the guide to choose desired objects to cut/engrave.

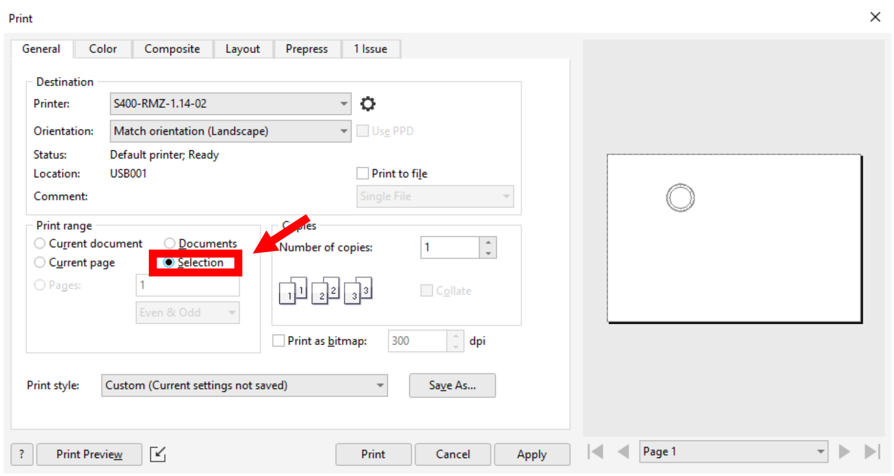
I. On the menu bar, Select the ***Print logo***.



II. Select ***"Current document"*** to choose all objects in the file

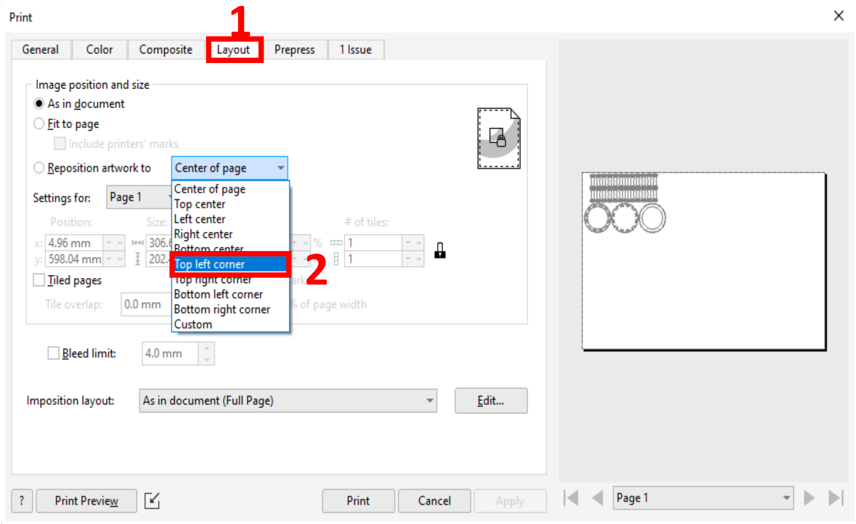


or ***"Selection"*** for specific object(s).



6.3. Layout

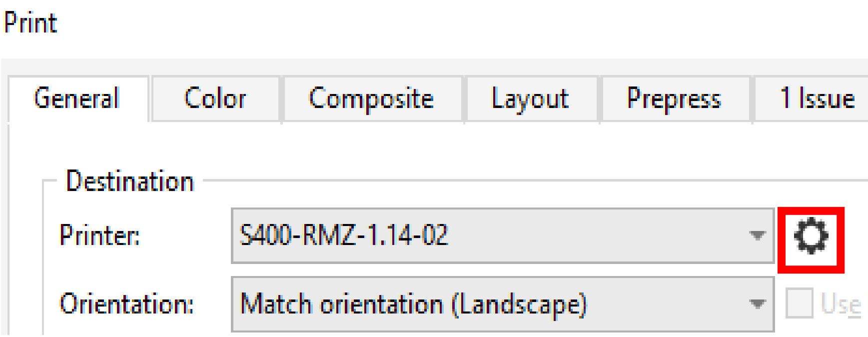
In ***"Print"***, select ***"Layout"***. Also, select ***"Top left corner"*** to ensure that the object.



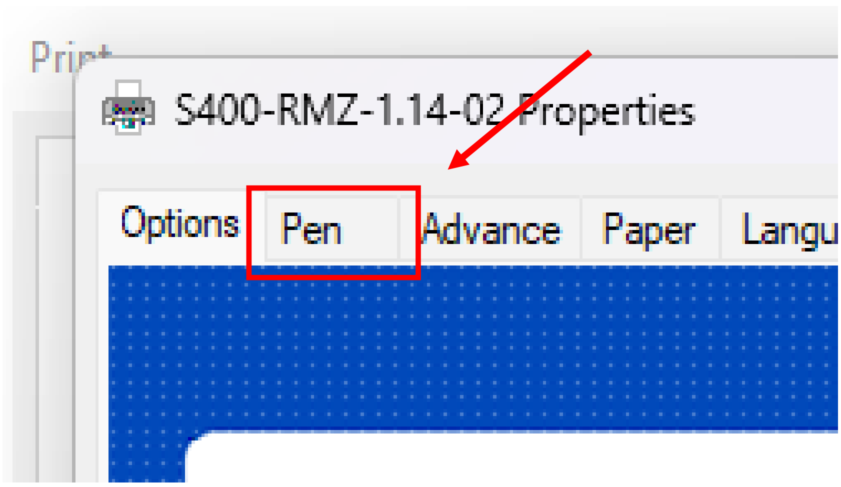
6.4. Pen & line color

This function is to specify the action of cutting / engraving and the order of cutting / engraving.

- Click the ***GEAR*** next to ***"Printer"***.

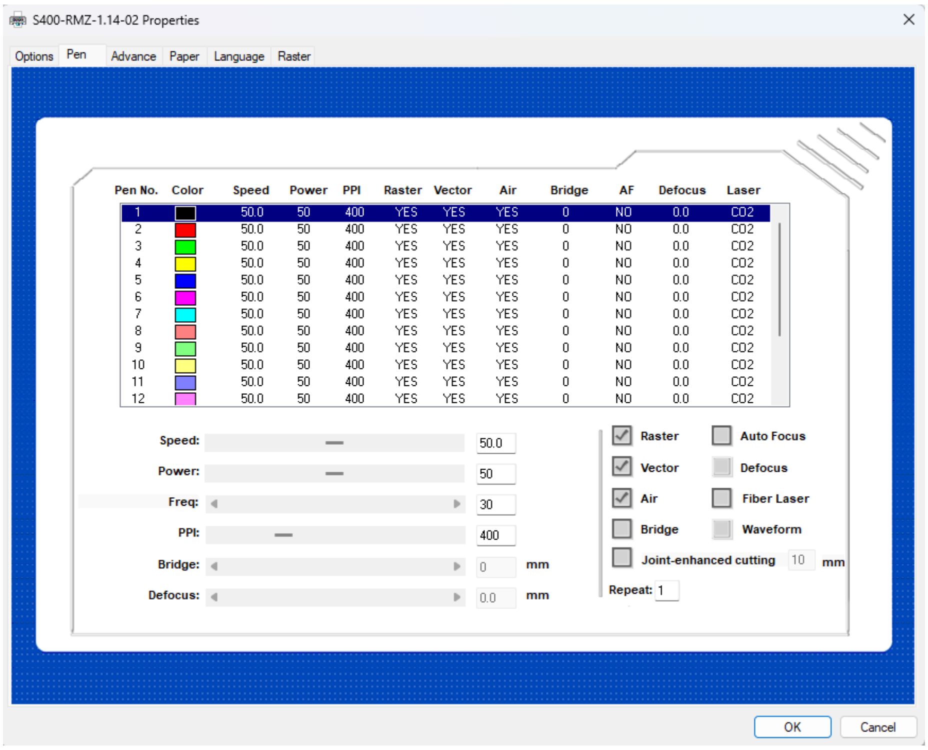


- Click ***"Pen ".***



Interpretation:
- ***"Pen No."***: Specify the cutting / engraving **sequence**
- ***"Speed"***: Specifies the **travelling speed** of laser head.
- ***"Power"***: Specifies the **power of laser** beam.

(Note: Different sets of ***"Speed" & "Power"*** settings decides which actions to be taken.)



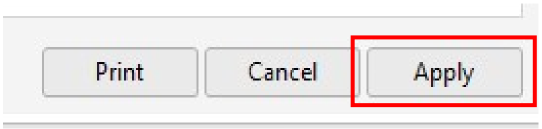
Recommended settings:

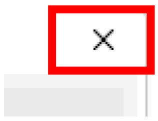
Action	Color	Speed	Power	Result
Light Engraving	**Black**	50	10	
Deep Engraving	**Green**	50	25	
Cutting	**Red**	2.2	60	

(Note: Engraving **MUST** be done **BEFORE** Cutting. Otherwise, the workpiece may be dislocated thus cause the engraving pattern to dislocate.)

- Click ***"OK"*** after finish altering the settings.

- Click the "Apply" and Cross.

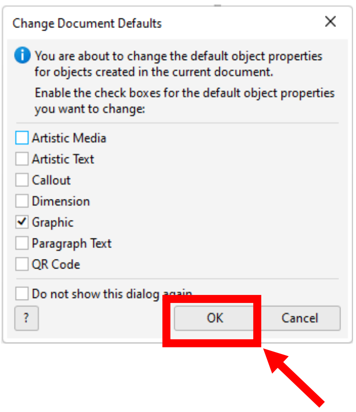




Next, assign different colors to the lines/ pattern.

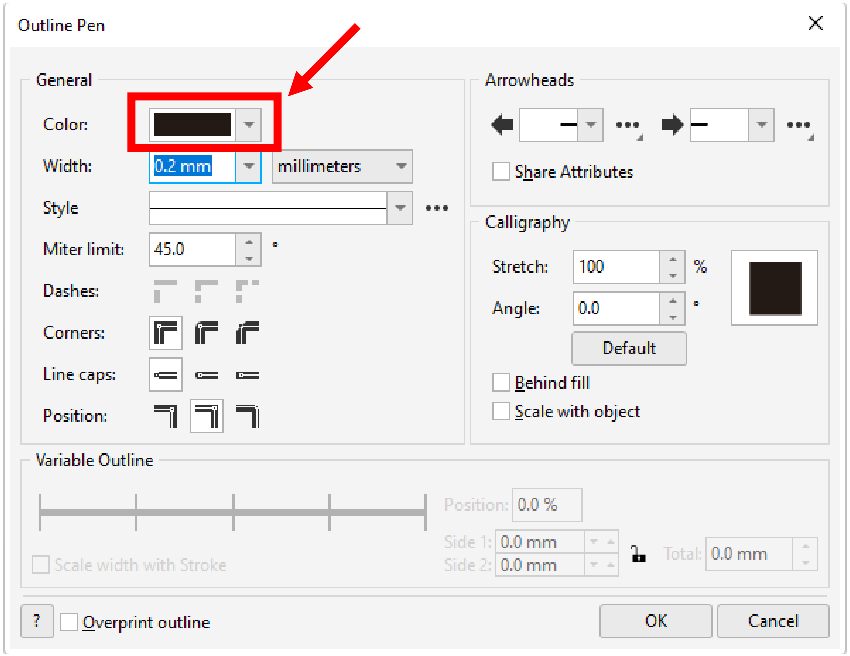
- On the bottom right corner, ***double-click*** this ***outline symbol.***

- Click ***"OK"***.

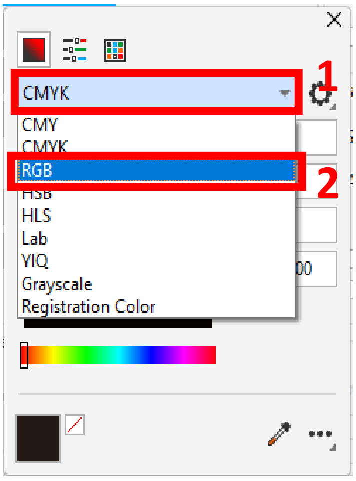


This page will be displayed.

- Click ***"Color".***



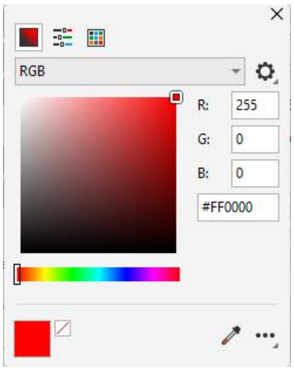
- Change ***"CMYK"*** to ***"RGB".***



Add the colors chosen in the ***"Pen"*** settings. Make sure they match each other.
Below is the RGB settings of the commonly used colors.

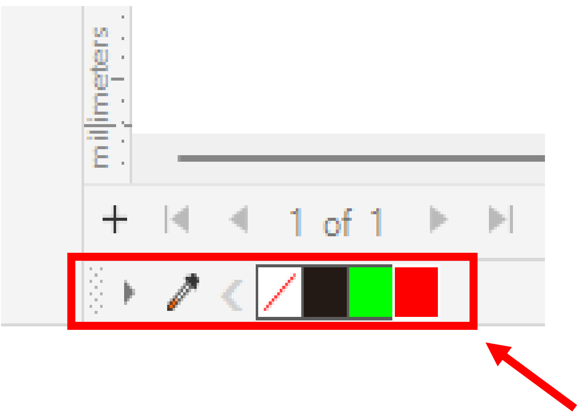
Color	Black	Red	Green
R	0	255	0
G	0	0	255
B	0	0	0

This is an example of the **Red** Color setting.

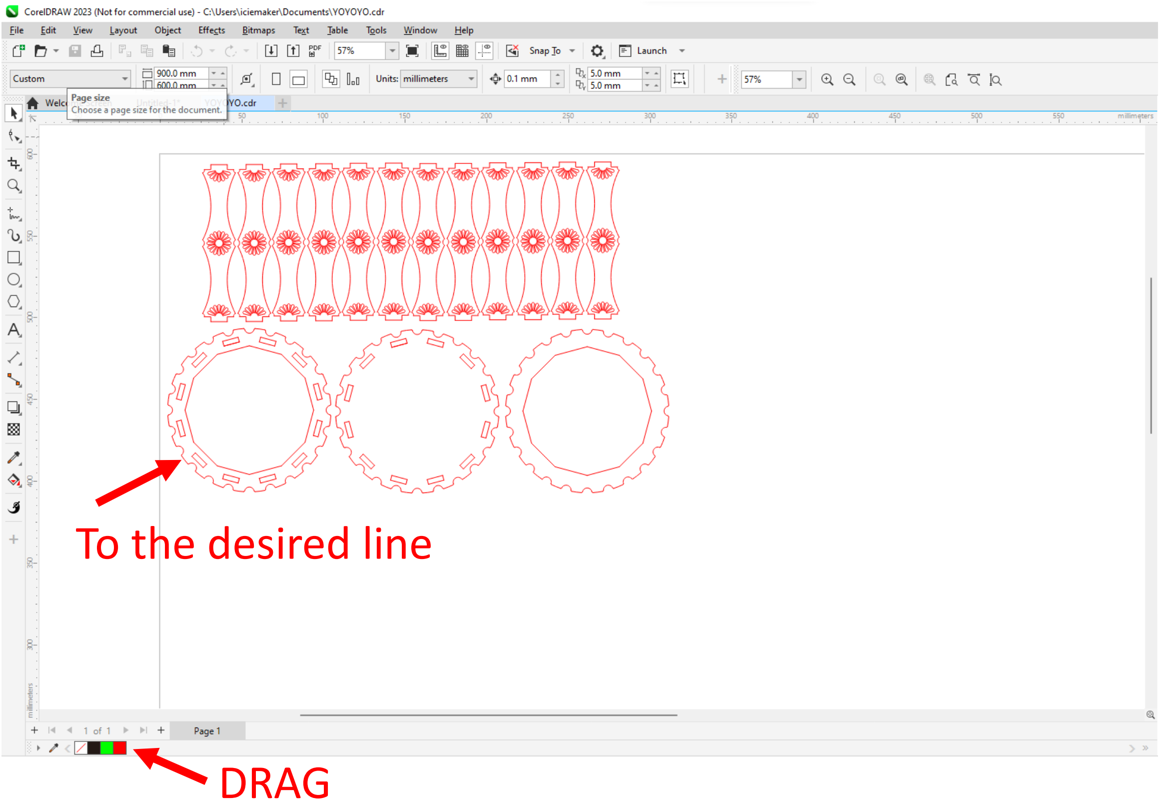


- Click ***"OK"*** after finish.

There are a few color blocks displayed on the bottom left corner. They are the colors just added.



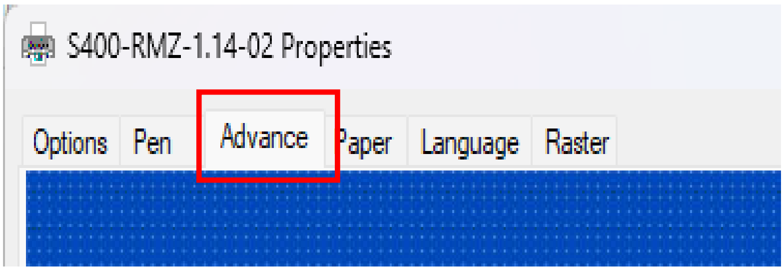
Drag the color blocks onto the desired line(s).



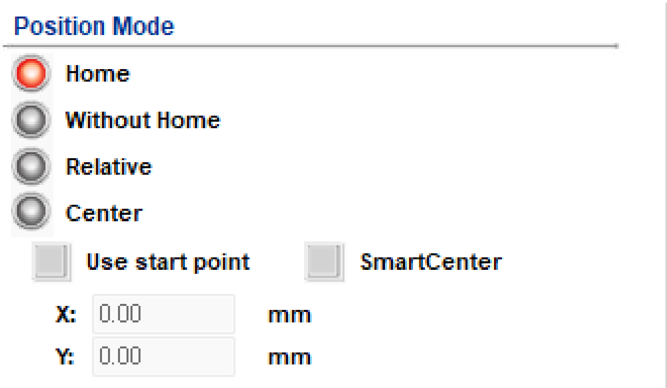
6.5. Advance

Go to ***"Print" > "Gear"*** again.

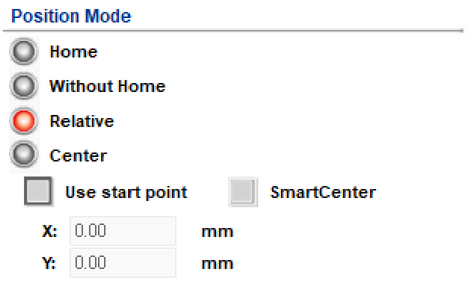
- Select "Advance".



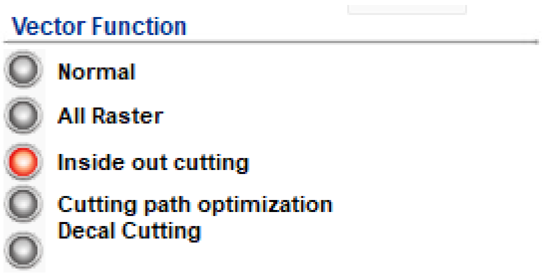
- Under Position mode, choose “Home” to start cutting at the home position (Top Left Corner).



or “Relative” to start cutting at a *desired position.*



- Under “Vector Function”, Choose ***“Inside out cutting”*** to ensure that the cutting begins from the “inside” features.



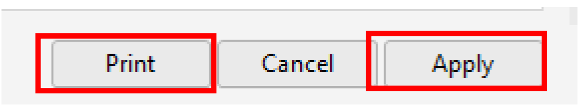
(Note: If the object / pattern is too complicated, it may not be able to distinguish “inside” and “outside” objects / pattern. In this context, manually assign colors to the object / pattern to ensure the work follow the desired sequence. (Refer to 6.4 Pen & line color))

- Click ***“OK”*** after finish.

Suggestion: **DOUBLE CHECK** the settings.

Step 7. Print file

After finish checking, click “Apply” and “Print”. The file will be transferred to the laser cut machine.



The file number will be increased by 1 if the file is successful transferred.

Start Cutting:

Step 1. Trial run

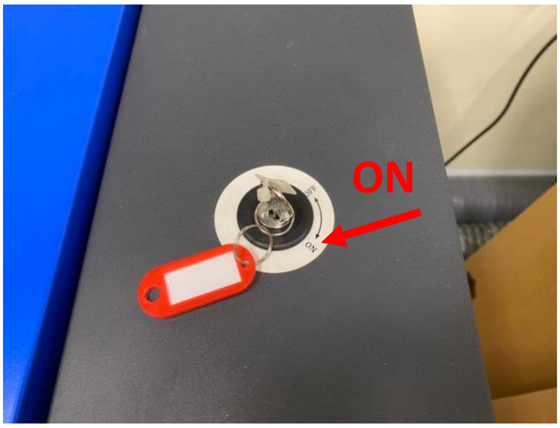
This step is important to see if the machine work in the desired path.

Check whether you have <u>removed the auto focus tool</u> again and <u>remain the key in "OFF" position</u>!!!

Tap "START/STOP" to start the trial run.

Step 2. Real trial

If the path is correct, you can start the cut.
Turn the key to ***"ON"*** position and tap ***"START/STOP"***.



Note:
If you want to **stop the job permanently, tap "STOP".**
If you want to **stop the job for a while and resume the job later, tap "Pause".**

If you encounter any emergencies, push the Emergency Stop.

E. Post-cutting:

Step 1. Get the product

After finish cutting, you may get the product. Remember to take the excess materials out.

Step 2. Return the key

Remove and return the key.

Step 3. Switch off all devices

Switch off the S400 and the ventilation system before leaving!